

Nunna Mokshith

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SUMMARY

B.Tech student in Robotics and Artificial Intelligence at Amrita Vishwa Vidyapeetham and a dedicated researcher with peer-reviewed work and a finalist entry in the IEEE IES 2026 Challenge. Experienced in Physics-Informed Machine Learning (PIML), systems automation, and data science pipelines, with a proven track record of architecting industrial condition diagnostics and clinical-grade biometric monitoring systems. Developer of *Wellsy v1.0*, an offline voice-interactive AI desktop companion and multi-threaded C++ hardware tools. Co-architect of *Dynamic Diffusion*, a generative AI-based zero-shot chemical synthesis platform. A dedicated athlete with strong leadership, disciplined teamwork, and effective communication skills in high-stakes environments. Currently seeking internship opportunities in Agentic AI and advanced AI agent architecture development.

EDUCATION

Bachelor of Technology (B.Tech) in Robotics and Artificial Intelligence , Amrita Vishwa Vidyapeetham CGPA : 8.96	08/2024 – Present Amritapuri
Higher Secondary Education , Narayana E-techno Junior College Board of secondary education - Percentage : 94.7%	07/2022 – 03/2024 Tirupathi

SKILLS

Embedded & Robotics ESP8266-NodeMCU, Arduino UNO, GSM/LTE (A7670C), Wi-Fi, I2C, MAX30102, Ultrasonic Sensors.	AI & Automation Agentic workflows Design and Management, LangGraph, RAG Pipelines, FastAPI, WebSockets. Antigravity, MSI integration	Machine Learning Physics-Informed Machine Learning (PIML), Graph Neural Networks (GNN), Physics-Informed Neural Operators (PINO), Predictive Maintenance, Exploratory Data Analysis (EDA), Data Normalization, Optuna, SMOTE, SHAP (Explainable AI)	Soft Skills Problem Solving Leadership Skills and Teamwork Project Management and Research Adaptability
Programming C, C++, Python , Arduino (C++)	Design & Tools AutoCAD, Git/GitHub, Geany.		

CERTIFICATES

Google Cloud Computing Foundations: Cloud Computing Fundamentals Google skills June 2026	Deploy and Manage Generative AI Models Google skills June 2026	Machine Learning for Beginners Simplilearn May 2026	Data Science Course - Mastering the Fundamentals Scaler April 2026
Guidewire DEVTrails University Hackathon March 2026	Quantum Computing: IBM Quantum Qiskit November 2025	Using Raspberry Pi & OpenCV Infosys Springboard November 2025	Developing Interpersonal Skills IBM (via Coursera) October 2025

PROJECTS

Wellsy v1.0: Asynchronous Desktop AI Agent Engine Engineered a native, fully offline-capable personal AI desktop assistant utilizing an asynchronous FastAPI WebSocket server backend communicating continuously with an Electron.js frontend dashboard. Developed a multi-node state machine engine using LangGraph and LangChain to govern a continuous Reasoning and Acting (ReAct) agent architecture, integrating a custom Model Context Protocol (MCP) server for standardized tool execution and local vector embedding RAG document indexing. Programmed structural multi-threaded native C++ file-scanning backend tool components and voice-native dialog handling modules using local speech-to-text structures to ensure high-speed, zero-latency system orchestration.	07/2026
Physics-Informed Multi-Modal Industrial Diagnostic Framework Architected a novel Physics-Informed Machine Learning (PIML) framework by merging data science with mechanical heuristics. By deriving six physical attributes—including True Mechanical Power, Specific Thermal Load, and Torque Shock—I integrated thermodynamic and kinetic laws to overcome sensor noise. To address extreme class imbalance, I implemented the SMOTE algorithm and a Custom Hybrid Ensemble, achieving an 85.2% recall for catastrophic failure detection. The system utilizes SHAP-based explainability for intrinsic interpretability, simulating a 68.0% reduction in maintenance costs across 2,000 devices.	05/2026
Physics-Guided Diffusion with Thermodynamic Awareness for 3D Molecular Generation Developed a generative framework utilizing Physics-Informed Neural Operators (PINO), qualifying as a Finalist (Milestone 3) in the IEEE IES 2026 challenge. I engineered Agentic AI workflows using Claude and Antigravity to automate model iteration and complex simulations. The architecture uses a dual-head GNN to optimize for geometric denoising and thermodynamic stability via a PINO-style Gibbs free-energy loss. This approach achieved 89.5% structural validity (a 15.5% baseline improvement) and reduced Gibbs Mean Absolute Error (MAE) by 63%. The system generated 250 unique molecules in 62.5s with 76.5% Lipinski compliance, utilizing a 3-stage RDKit cascade for chemical validation.	03/2026
Hybrid GSM & Wi-Fi Smart Switch for Industrial Automation , Prof. Sumesh K.P Designed and engineered a robust remote-automation node under the guidance of Prof. Sumesh K.P to manage industrial loads in off-grid locations. By integrating an ESP8266 NodeMCU with an A7670C module, I implemented a dual-layer connectivity system that ensures 100% uptime through a GSM/Wi-Fi failover mechanism. The core of the system utilizes a fail-safe asynchronous parser to handle SMS-based interrupts, providing a reliable solution for remote hardware resets and high-fidelity telemetry with zero internet dependency.	12/2025

RESEARCH & PUBLICATIONS

Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity International Conference on Robotics and Mechatronics (ICRM)	11/2025
Autonomous Architecture: Developed a Raspberry Pi-based platform to emulate Level 3-4 autonomous driving functionalities. Perception Systems: Integrated a multi-sensor suite for real-time obstacle avoidance and computer vision-based environment perception. System Optimization: Applied control theory and machine learning principles to streamline decision-making in complex driving scenarios	